

HERO scorecard (vertical) — value ± BCa 95% CI · colour = per-metric min-max (green = best) · * = Δ-vs-frozen CI excludes 0
13 metrics × 20 encoders · WINNER col + BLUE BOX = surgery-vs-vanilla-cont-SSL champion duel (tie = paired 95% CI overlaps 0) · 10 K bootstrap

	G-frozen	G-fullFT	G-LPFT	G-DoRA	G-LoRA	G-vCSSL-enc	G-raw	G-surgical	G-s3DI-enc	G-s3DI-hd	G-surgical	G-surgical	G-surgical	G-surgical	G-argn	G-surgical	G-surgical	G-surgical	G-sNoDI-enc	G-sNoDI-hd	WINNER
action_top1 ↑ better	44.274 ±2.274	52.712 ±2.301*	49.041 ±2.329*	52.603 ±2.329*	50.904 ±2.274*	52.822 ±2.301*	50.685 ±2.301*	50.027 ±2.247*	49.151 ±2.320*	49.973 ±2.301*	50.849 ±2.285*	50.027 ±2.329*	49.425 ±2.301*	48.548 ±2.274*	52.274 ±2.301*	48.164 ±2.274*	47.562 ±2.274*	47.452 ±2.274*	50.575 ±2.301*	49.644 ±2.274*	tie
motion_cos ↑ better	0.009 ±0.001	0.244 ±0.006*	0.119 ±0.004*	0.161 ±0.005*	0.170 ±0.005*	0.151 ±0.004*	0.133 ±0.004*	0.116 ±0.004*	0.119 ±0.004*	0.127 ±0.005*	0.121 ±0.004*	0.117 ±0.004*	0.118 ±0.004*	0.062 ±0.002*	0.189 ±0.005*	0.062 ±0.002*	0.035 ±0.001*	0.018 ±0.001*	0.127 ±0.004*	0.135 ±0.005*	ViT-G surgical autorgn encoder
taxonomy_f1 ↑ better	0.797 ±0.017	0.785 ±0.017	0.789 ±0.017	0.782 ±0.017	0.786 ±0.017	0.790 ±0.017	0.792 ±0.017	0.789 ±0.017	0.789 ±0.017	0.787 ±0.017	0.787 ±0.017	0.792 ±0.017	0.790 ±0.017	0.792 ±0.017	0.779 ±0.017*	0.793 ±0.017	0.794 ±0.017	0.801 ±0.016	0.785 ±0.017	0.787 ±0.017	tie
future_mse ↓ better	0.557 ±0.001	0.541 ±0.001*	0.536 ±0.001*	0.514 ±0.001*	0.504 ±0.001*	0.533 ±0.001*	0.497 ±0.001*	0.496 ±0.001*	0.498 ±0.001*	0.533 ±0.001*	0.496 ±0.001*	0.497 ±0.001*	0.496 ±0.001*	N/A	0.526 ±0.001*	0.508 ±0.001*	0.519 ±0.001*	0.533 ±0.001*	0.498 ±0.001*	0.533 ±0.001*	ViT-G surgical 3stage DI intervene encoder
rollout ↓ better	0.009 ±0.000	0.010 ±0.000*	0.010 ±0.000*	0.011 ±0.000*	0.009 ±0.000*	0.008 ±0.000*	0.010 ±0.000*	0.011 ±0.000*	0.010 ±0.000*	0.008 ±0.000*	0.010 ±0.000*	0.010 ±0.000*	0.012 ±0.000*	N/A	0.009 ±0.000*	0.009 ±0.000*	0.009 ±0.000	0.008 ±0.000*	0.012 ±0.000*	0.008 ±0.000*	tie
causal ↓ better	0.583 ±0.001	0.566 ±0.001*	0.561 ±0.001*	0.542 ±0.001*	0.533 ±0.001*	0.558 ±0.001*	0.528 ±0.002*	0.529 ±0.002*	0.530 ±0.001*	0.558 ±0.001*	0.528 ±0.002*	0.531 ±0.002*	0.529 ±0.002*	N/A	0.551 ±0.001*	0.540 ±0.001*	0.550 ±0.001*	0.561 ±0.001*	0.530 ±0.002*	0.558 ±0.001*	ViT-G surgery raw encoder
tdist ↓ better	0.007 ±0.000	0.008 ±0.000*	0.008 ±0.000*	0.007 ±0.000*	0.008 ±0.000*	0.006 ±0.000*	0.008 ±0.000*	0.008 ±0.000*	0.007 ±0.000*	0.006 ±0.000*	0.008 ±0.000*	0.008 ±0.000*	0.008 ±0.000*	N/A	0.008 ±0.000*	0.007 ±0.000*	0.007 ±0.000	0.007 ±0.000	0.008 ±0.000*	0.006 ±0.000*	tie
teacher_free ↓ better	0.046 ±0.000	0.049 ±0.001*	0.048 ±0.001*	0.051 ±0.001*	0.045 ±0.001	0.038 ±0.001*	0.045 ±0.001*	0.048 ±0.001*	0.045 ±0.001	0.038 ±0.001*	0.047 ±0.001*	0.047 ±0.001*	0.052 ±0.001*	N/A	0.047 ±0.001*	0.044 ±0.001*	0.044 ±0.001*	0.043 ±0.001*	0.052 ±0.001*	0.038 ±0.001*	tie
maskratio ↓ better	0.072 ±0.001	0.066 ±0.001*	0.061 ±0.001*	0.057 ±0.000*	0.054 ±0.000*	0.063 ±0.001*	0.056 ±0.001*	0.053 ±0.000*	0.057 ±0.000*	0.063 ±0.001*	0.060 ±0.001*	0.057 ±0.001*	0.054 ±0.000*	N/A	0.061 ±0.001*	0.064 ±0.001*	0.066 ±0.001*	0.070 ±0.001*	0.057 ±0.001*	0.063 ±0.001*	ViT-G surgical 3stage DI diheavy encoder
aot ↑ better	0.963 ±0.006	0.935 ±0.009*	0.946 ±0.008*	0.941 ±0.008*	0.938 ±0.008*	0.946 ±0.008*	0.935 ±0.009*	0.935 ±0.008*	0.932 ±0.009*	N/A	0.934 ±0.009*	0.934 ±0.008*	0.942 ±0.008*	0.956 ±0.007*	0.941 ±0.008*	0.954 ±0.007*	0.959 ±0.007	0.964 ±0.007	0.940 ±0.008*	N/A	ViT-G surgical intervene wiseft f70 encoder
tov ↑ better	0.950 ±0.005	0.932 ±0.006*	0.942 ±0.006*	0.937 ±0.006*	0.926 ±0.006*	0.928 ±0.007*	0.937 ±0.006*	0.944 ±0.006*	0.943 ±0.006*	N/A	0.943 ±0.006*	0.941 ±0.006*	0.930 ±0.006*	0.941 ±0.006*	0.943 ±0.006*	0.946 ±0.005	0.950 ±0.005	0.953 ±0.005	0.942 ±0.006*	N/A	ViT-G surgical intervene wiseft f70 encoder
pace ↑ better	0.741 ±0.013	0.717 ±0.012*	0.708 ±0.012*	0.717 ±0.013*	0.724 ±0.013*	0.731 ±0.013*	0.730 ±0.013*	0.719 ±0.012*	0.731 ±0.013*	N/A	0.729 ±0.013*	0.724 ±0.013*	0.717 ±0.012*	0.704 ±0.012*	0.722 ±0.013*	0.736 ±0.013	0.725 ±0.013*	0.727 ±0.013*	0.712 ±0.012*	N/A	tie
tcc_tau ↑ better	0.293 ±0.005	0.284 ±0.005*	0.260 ±0.005*	0.269 ±0.005*	0.270 ±0.005*	0.292 ±0.006	0.252 ±0.005*	0.274 ±0.006*	0.269 ±0.006*	N/A	0.264 ±0.005*	0.270 ±0.005*	0.268 ±0.005*	0.278 ±0.006*	0.269 ±0.005*	0.274 ±0.006*	0.280 ±0.005*	0.288 ±0.005*	0.262 ±0.005*	N/A	ViT-G vanilla continual-SSL